Sahaj Bhadja

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Summary

Software engineer with 2+ years building **C/C++, Python, cross-platform systems, and autonomous-driving simulations**; experienced in operating systems, testing, debugging, system integration, technical documentation, reliability, and healthcare application development.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Object-Oriented Programming, Software Engineering, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Designed and tested **fault-tolerant Python system modules** with explicit states, failure isolation, and validation, processing 1,000+ concurrent jobs while improving throughput 3x and preventing cascading production failures.
- Diagnosed complex software and data failures through targeted tests, logs, and performance analysis, sustaining 99%+ reliability across 5,000+ recurring monthly workflows while reducing database latency 35%.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Built and debugged a **multithreaded Python processing engine** using ThreadPoolExecutor, secure file ingestion, and fault-tolerant execution, handling 1,000+ files per batch with predictable, stable peak-load behavior.
- Owned software modules through requirements, implementation, unit and integration testing, code review, deployment, and detailed documentation, reducing feedback turnaround 60% while supporting 10,000+ daily production requests.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*

- Developed **autonomous-driving simulation and data pipelines** over 100,000+ trajectories, translating system requirements into reproducible Python experiments that improved long-horizon behavioral planning performance 18% under controlled evaluations.
- Designed repeatable system-level tests across 10,000 driving episodes, analyzing safety-relevant failures, road geometry, and closed-loop behavior to improve planning stability 15% while documenting experimental design decisions.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Developed and debugged **cross-platform Java and Python modules** using OOP patterns, APIs, SQL, and caching across Linux, Windows, and macOS, reducing response times 30% and technical debt 20%.
- Automated builds, installations, and 20+ unit and integration tests with Jenkins, PyTest, Mockito, and Git, producing detailed traceable failure reports while collaborating with globally distributed engineering teams.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Integrated **eight modular application services** for authentication, records, permissions, billing, analytics, and production operational monitoring workflows, securely synchronizing 2M+ records through REST APIs, WebSockets, and Azure SQL.
- Supported automated verification and system testing through Jest, Selenium, Jenkins, and production observability, achieving 95%+ build validation while reducing deployment time 50% through Docker-based repeatable delivery workflows.

Skills

Programming: C, C++, Python, Java, Bash, object-oriented programming, data structures, algorithms
Systems: Operating systems, Linux, concurrency, multithreading, system integration, distributed systems, real-time data, performance analysis
Quality: Unit testing, integration testing, system-level testing, verification, validation, debugging, failure isolation, root-cause analysis
Tools: Git, Jenkins, GitHub Actions, PyTest, Mockito, Jest, Selenium, Docker, CI/CD, logging, monitoring
Documentation: Software requirements, design decisions, test results, technical documentation, code reviews, maintainable modules, Agile collaboration

Projects

CPU Scheduling Simulator Live | *C++, React.js, Operating Systems*

- Implemented and verified **six C++ scheduling algorithms** with interactive controls, Gantt visualization, comprehensive edge-case testing, and dynamic metrics across 1,000+ process combinations, reducing average turnaround time 25%.

HMIS Live | *TypeScript, REST APIs, Sentry*

- Built a **secure healthcare workflow application** with consent-based access, provider dashboards, appointments, REST integrations, notifications, and system monitoring, supporting 100+ daily workflows while reducing administrative workload 30%.

Chat2SQL Code | *Java, TypeScript, Docker, Linux*

- Enhanced an **existing cross-platform database application** across Java and TypeScript modules, supporting 10+ database systems with structured execution, validation, Docker, Maven, Linux-compatible builds, and reusable debugging workflows.